PICO 2 / PICO9918 BENCH PROTOTYPE

Direct bus + SWD | 40-pin ZIF | Manual power | 12 September 2026

3.3 V data writes: confirm target U5 is HCT245. Saved schematics say HC245; direct writes then need bench validation.
One USB source powers both boards through Pico 2 VBUS. GP26-GP28 are NOT 5 V tolerant; no bus wires use them.

PROBE: PICO 2 HEADER

TARGET: ZIF SOCKET

GPIO	Physical pin		DIP pin	Target header
GP4	6	← **D0 / CD7** →	**17**	H2.7
GP5	7	← **D1 / CD6** →	**18**	H2.8
GP6	9	← **D2 / CD5** →	**19**	H2.9
GP7	10	← **D3 / CD4** →	**20**	H2.10
GP8	11	← **D4 / CD3** →	**21**	H3.1
GP9	12	← **D5 / CD2** →	**22**	H3.2
GP10	14	← **D6 / CD1** →	**23**	H3.3
GP11	15	← **D7 / CD0** →	**24**	H3.4
GP12	16	— **/CSR** →	**15**	H2.5
GP13	17	— **/CSW** →	**14**	H2.4
GP14	19	— **MODE** →	**13**	H2.3
GP15	20	— **MODE1** →	**11**	H2.1
GP21	27	— **/RST** →	**34**	H4.2
GP18	24	← **/INT** —	**16**	H2.6
GP19	25	← **CPUCLK** —	**38**	H4.6
GP20	26	← **GROMCLK** —	**37**	H4.5
GND	23	— **GND** —	**12**	H2.2

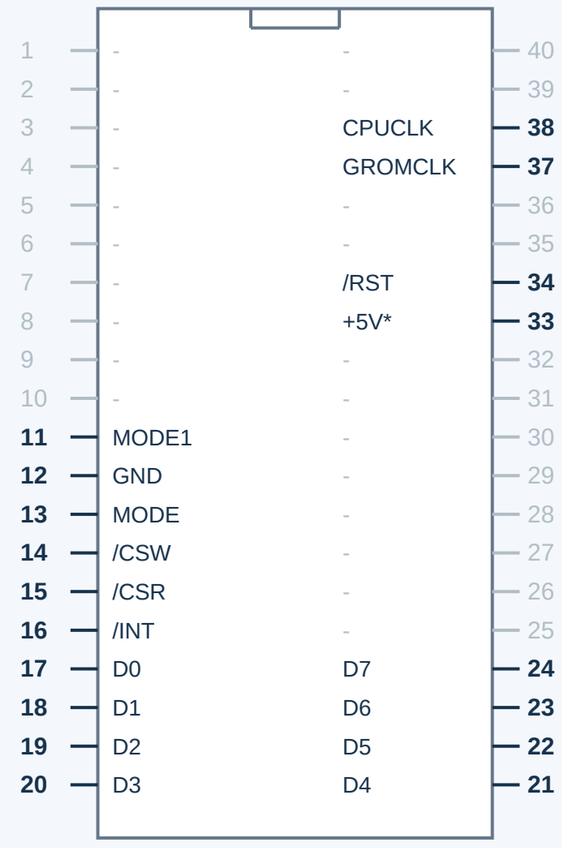
Arrows show signal direction; data pins switch between input and output in firmware.

SWD: SEPARATE TARGET DEBUG CONNECTOR

GP2 / pin 4	— **SWCLK** →	DBG1 pin 1
GND / pin 3	— **GND** —	DBG1 pin 2
GP3 / pin 5	← **SWDIO** →	DBG1 pin 3

ZIF: TOP / INSERTION VIEW

Align target pin 1 with socket pin 1.



* Pin 33: Pico VBUS via jumper or page-2 switch.

15.24 mm target rows; socket body not to scale.

POWER + VIDEO

PC USB -> Pico 2 USB; VBUS pin 40 -> removable jumper -> ZIF pin 33. Leave target USB unplugged.
Target video FFC -> existing VGA adapter -> monitor. No video signals go through the probe.

1. Configure /CSR, /CSW, MODE, MODE1 and /RST idle HIGH, then fit the target power jumper.
2. Make GP4-GP11 inputs BEFORE asserting /CSR. Release /CSR and allow bus turnaround before driving data.
3. For a manual power cycle, release bus/SWD outputs and pulls, remove the power jumper, then reconnect.

Manual option shown. Optional transistor power switch: see page 2. Use GP2/GP3, not the probe native SWD pads.